Home Automation Controller Unit

Technical Specification and Interface Datasheet

Document ID DS-HAC-2026-V1 | Version 1.0 | October 2026

1 System overview and product highlights

The Home Automation Controller Unit provides four relay channels for programmable household lighting and compatible device control. A local manual switch input allows on and off override, while a UART programming interface supports firmware updates and custom control logic. This datasheet describes the power connections, relay outputs, switch input, and installation requirements.

- Four normally open relay channels labeled NO1, NO2, NO3, and NO4 for independently configured household loads.
- DC supply connections through a VCC screw terminal or a nominal 12 VDC XT60 connector.
- Active-low IO27 input for a momentary wall or panel switch that toggles manual on and off states.
- RESET and BOOT buttons with a UART header for programming and diagnostics.
- Flanged ABS housing and a protective PCB conformal coating as described for the controller hardware.

2 Home automation applications

Channel	Example application
NO1	Room or hallway lighting
NO2	Outdoor or porch lighting within a suitable enclosure
NO3	A compatible ventilation fan or a properly rated contactor control circuit
NO4	Auxiliary low-voltage lighting or another compatible household load

Channel assignments and automation rules depend on the installed firmware. Time schedules, sensor control, Wi-Fi, Bluetooth, smartphone access, and smart home platform integration require suitable firmware and hardware; they are not established by the interface specifications alone.

3 Physical and environmental characteristics

Housing: flanged ABS enclosure. PCB protection: conformal coating. Dimensions, mounting-hole spacing, operating temperature, humidity range, and enclosure ingress-protection rating must be confirmed for the installation. Outdoor or damp-location suitability is not specified.

4 Electrical specifications

The following values retain the stated controller specifications. Relay contact ratings are component limits and do not establish the complete assembly rating for household mains loads.

Parameter	Stated value	Conditions or limits
VCC screw terminal	12 VDC minimum; 12 or 24 VDC nominal; 35 VDC maximum	35 VDC is the stated absolute maximum; use a regulated nominal supply with margin.
XT60 input	12 VDC nominal; 14.2 VDC maximum	Dedicated 12 VDC input. Do not apply the VCC terminal maximum to XT60.
Relay contacts	250 VAC / 30 VDC; 10 A maximum	Normally open dry contacts. Confirm load type, inrush, terminal capacity, and assembly rating.
IO27 idle level	3.0 to 3.3 V; nominal 3.3 V	Pulled high; stated active-low threshold below 0.8 V.
IO27 current to GND	0.33 mA nominal; 1.0 mA maximum	Stated current when input is pulled low.
UART logic	3.3 V recommended pending verification	The original specifies 3.3/5 V TTL compatibility; 5 V tolerance must be confirmed before use.
Idle supply current	15 to 75 mA; nominal 35 mA	At 12 VDC with all relays de-energized.
Supply current with all relays on	120 to 220 mA; nominal 160 mA	At 12 VDC; excludes external switched-load current.

5 Relay load selection

Load category	Stated limit	Home installation guidance
DC resistive loads	12 to 30 VDC; up to 10 A	Use a fused supply and verify conductor and terminal capacity.
AC loads	110 to 250 VAC; up to 10 A	Use only after confirming complete-board mains suitability, insulation, and enclosure requirements.
Inductive loads	12 or 24 VDC; up to 5 A	Verify inrush and suppression. A relay current rating alone does not establish motor compatibility.

The stated inductive-load limit does not establish an AC motor rating. Use an appropriately rated contactor where required. Confirm whether load suppression is actually fitted; the controller description does not establish protection for every external load.

6 Terminal and header descriptions

Verify the printed terminal labels and board wiring before connecting power or loads. Keep low-voltage logic wiring separated from mains wiring.

Location	Label	Function
Left terminal block	IO27	Active-low manual input. Connect a momentary dry-contact switch between IO27 and GND.
Left terminal block	GND x2	Common controller ground terminals for the DC supply return and switch reference.
Left terminal block	VCC	Main DC supply input. Stated range: 12 to 35 VDC maximum. Reverse-polarity protection is described; verify before relying on it.
Power socket	12VDC XT60	Polarized nominal 12 VDC power input; observe the separate 14.2 VDC maximum.
Right terminal block	NO4	Normally open relay contact for channel 4.
Right terminal block	NO3	Normally open relay contact for channel 3.
Right terminal block	NO2	Normally open relay contact for channel 2.
Right terminal block	NO1	Normally open relay contact for channel 1.
Programming header	UART	Six-pin header described as providing TX, RX, GND, and 3V3. Verify physical pin order and the remaining pins.
Onboard control	RESET	Momentary button for a hardware processor reset.
Onboard control	BOOT	Button used with reset or power-up to enter the serial bootloader.

Relay common connection: identify the corresponding COM terminal or documented shared common for each NO output. The available labels do not define the common-contact topology. Do not assume any relay common is internally connected to VCC or that all relay channels have separate commons.

7 Manual override behavior

- Idle: IO27 stays high at approximately 3.3 V, and the controller runs its configured home automation program.
- Switch press: connecting IO27 to GND produces a falling edge that toggles the configured manual on and manual off states.
- Debounce: the stated firmware debounce interval is 50 ms. Verify this behavior in the installed firmware.

The affected channels, power-up state, and method for returning to automatic operation depend on firmware configuration. Confirm these behaviors before installation. The IO27 manual input is a control command; use a separate power disconnect for isolation.

8 Firmware flashing and programming

Use a USB-to-TTL adapter configured for verified controller logic levels. Firmware must match the actual processor and board configuration.

1. Disconnect switched loads and make the programming area safe. Identify TX, RX, and GND on the UART header.

2. Connect adapter RX to controller TX, adapter TX to controller RX, and adapter GND to controller GND. Use 3.3 V logic unless 5 V tolerance has been confirmed.

3. Power the controller from an appropriate supply. Do not connect the adapter power pin unless the board power arrangement has been verified.

4. Press and hold BOOT. While holding BOOT, press and release RESET, then release BOOT.

5. Start the flashing utility with the correct firmware and processor settings. The stated programming baud rate is 115200.

6. Reset the controller and test each relay, the manual switch, and the configured automation behavior before reconnecting loads.

9 Application and wiring guidelines

- Use a suitable regulated DC supply and input fuse. Select wire size for current, length, voltage drop, and terminal capacity.
- Use one power input at a time unless the board documentation explicitly confirms safe simultaneous use of VCC and XT60.
- Connect a momentary dry-contact switch between IO27 and GND. Do not apply mains voltage or external supply voltage to IO27.
- Route the load supply through the verified relay COM and NO contacts. Dry contacts switch an external circuit; they do not inherently supply load power.
- Fit load-appropriate suppression for inductive devices after confirming the protection already present on the board.

Household mains connections require a suitably qualified installer, appropriate circuit protection, and an enclosure that prevents access to live parts. Confirm the complete assembly is suitable for the intended mains load before use.

10 Typical low voltage connection summary

Connection	Wiring path
Controller supply	DC supply positive to VCC; DC supply negative to GND. Alternatively use the nominal 12 VDC XT60 input.
Manual switch	IO27 to one switch contact; the other switch contact to GND.
Channel 1 load	Fused load supply positive to verified COM1; NO1 to load positive; load negative to load supply negative.
Other channels	Repeat the load circuit for NO2 to NO4 using their verified common connections and compatible loads.

This summary applies to low-voltage DC wiring. Confirm the actual common-contact arrangement before following the load paths.